Fiscal Year:	FY 2016	**Task Last Updated:**	FY 12/06/2016
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Molecular Biology of Growth and Cell Remodeling within the Spaceflight Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/01/2012	**End Date:**	06/30/2016
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 06/30/2016 per KSC information (Ed., 7/25/16) NOTE: End date is now 12/31/2015 per NSSC information; original end date 7/31/13 (Ed., 6/24/14) NOTE: End date is now 6/30/2014 per NSSC information; original end date 7/31/13 (Ed., 8/9/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa (University of Florida)		
Grant/Contract No.:	NNX12AN69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants experiencing spaceflight are quite normal in appearance but can exhibit growth habits distinctly different from plants on Earth. Historically the spaceflight-induced differences were difficult to dissect due to changing hardware and flight conditions. With the Advanced Biological Research System (ABRS) and a consistent International Space Station (ISS) orbital environment, it is now possible to confidently approach a dissection of the molecular biology guiding the altered growth of plants in spaceflight. This proposal involves experiments that specifically address growth and molecular changes in Arabidopsis that occur during spaceflight, bringing the molecular and genetic tools of Arabidopsis to bear on fundamental questions of root morphology, growth, and cell wall remodeling. During the Advanced Plant Experiments on Orbit - Transgenic Arabidopsis Gene Expression System (APEX-TAGES) experiments, we observed a remarkable number of gene expression changes during spaceflight that are associated with cell wall restructuring, especially in roots. We also observed, with the ABRS/GIS (GFP Imaging System) imaging system, genetically dependent changes in root growth direction, movement, and structure. The current experiments will employ the ABRS and the GIS to dissect and better understand the molecular changes guiding growth restructuring on orbit. In particular we hypothesize that spaceflight conditions engender a constant state of root cell wall modification during the cell file rotations that characterize spaceflight-induced root skewing. The specific experiments will involve mutants in root growth and cell wall signaling pathways identified in our previous spaceflight molecular studies, as well as new Green Fluorescent Protein biosensor lines also based on the genes identified in the previous flight experiments. The overall goal of our program is to understand fundamental molecular biological responses to extraterrestrial environments. This goal aligns with fundamental Space Biology goals to understand the impact of spaceflight on biological systems to better enable the exploration imperative. We have used molecular biology and genetically tagged plants as biological monitors of spaceflight and space-related environments with great science return, leading to the development of new biological and hardware tools to study the spaceflight response with even higher fidelity in a wider variety of spaceflight-related environments. The focus of the current proposal is to build on these data and other insights gained from the execution of the APEX-TAGES experiments in the ABRS /GIS, advancing science while also advancing the telemetric imaging hardware and biological experiment support for the ISS and future platforms Gaining insights into such mechanisms is recognized as fundamental within the decadal study and underpins answers to some of the biggest questions in spaceflight plant biology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To gain fundamental insight into how plants perceive and process the signals that control tropism and adaptive physiology in novel and extreme environments. It also highlights how the unique research environment of the ISS can address biological questions that cannot be answered on Earth where gravity would mask many of the underlying phenomena.
Task Progress:	Final Summary: The APEX03 experiment was a natural follow-on of the APEX01 experiment, and is an outstanding example of how the ISS can be utilized as an exploration laboratory to more science forward in a given field. In APEX01 we showed that plants responded to the spaceflight environment in a highly tissue-specific manner, and that cell wall remodeling played a large role in the physiological adaptation to spaceflight. Further, we refuted the paradigm set forth by Darwin in the 1880s that plants require gravity to initiate the root behavior called “skewing,” and fluorescent reporter genes suggested that there were changes in auxin signaling over time on orbit. In APEX03 we set forth to dissect the molecular changes over a developmental time course, utilize mutants in critical skewing and cell wall remodeling signal transduction pathways, and also use targeted fluorescent imaging of auxin-regulated genes in the specific cells of the root tip that guide root navigation. Although the transcriptome analyses are ongoing, preliminary data suggest that development age plays a substantial role in the physiological adaptation process Transcriptomes were first evaluated four days after germination, where there were numerous differentially expressed genes (compared to ground controls). A set of sister plants were evaluated 8 days after germination on orbit, and it was found that as the plants develop, the number of genes needed to satisfy the needs of the physiological adaptation process diminishes. In the on orbit microscopy experiments (utilizing the Light Microscopy module – the LMM) we found that auxin distributions in the gravity-sensing cells of the primary root are not gravity dependent, even though these distributions are key to gravity responses on Earth. It is therefore likely that the role of gravity is minimal compared to the inherent developmental cues guided by auxin to define the architecture of the root tip. Those developmental auxin gradients were co-opted then to be sensitive to gravity responses and it seems logical then that the auxin signaling system in roots was present in plants as a developmental system before evolving additional roles in gravity responses. These data actually changed aspects of a long held paradigm in plant biology that gravity is central to setting up the gradient of auxin distribution in a vertically growing root – called the “reverse fountain” model of auxin distribution. The term “vertical” carries a direct association with the gravity vector. Without the ability to remove gravity from the plant environment, there was no reason to eliminate the term from the vernacular or the force of gravity from the concept. Our results show that the reverse fountain is neither reverse nor a fountain, as these terms imply that gravity is involved in the reversal of liquid flow. Rather, the auxin flow of the primary root tip is fundamentally no different from the auxin flow systems that guide other parts of plant primary development, except that it has been further engaged in the gravity response.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
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